



Figure 1: GNS3 starting

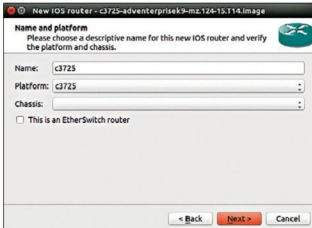


Figure 2: Adding new IOS router

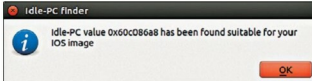


Figure 3: Suitable Idle-PC value

It uses a simple configuration file for specifying virtual router configurations, and enables interconnecting of routers and WAN technologies such as ATM, frame relays and switches. Further, it provides management CLI for device functions such as start, stop, suspend, reload, console connection, etc.

- Python: Dynagen is written in Python.
- VPCS: This is a virtual PC simulator, which can simulate up to nine PCs. It is possible to use the *ping* and *traceroute* commands from or towards these virtual PCs. Remember that VPCS is just a program running on Linux or Windows supporting a few network commands. But, it becomes difficult to configure a network without VPCS. Earlier, VPCS was not available, and users were required to configure an additional router in place of a host (PC) for testing.
- QEMU: A generic and open source machine emulator and virtualiser. As a machine emulator, it enables programs made for one system to be run on different machines (e.g., programs developed for ARM can be run on a PC).

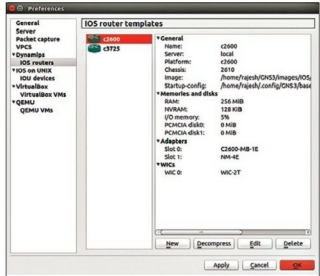


Figure 4: IOS router templates



Figure 5: Drag-and-drop router

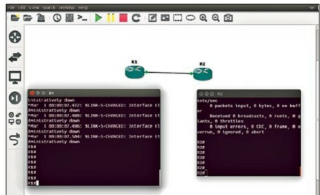


Figure 6: Console for two routers established

As a virtualiser, it achieves near native performances by executing the guest code directly on the host CPU.

- cpulimit: This attempts to limit the real CPU usage of a process. It is helpful to limit CPU usage of devices created in GNS3.
- VirtualBox: A virtualiser for x86 hardware, targeted at server, desktop and embedded use—a cross-platform virtualisation application.